Quiet Neurons: a BDH quietens on what it just learned, not on what is merely predictable

DataForge 2026, Pathway track · concept **Sparse Non-Negative Activations** · Nilay Toshniwal, solo · artifact **nilaymastaadmi.github.io/quiet-neurons** · code **github.com/nilaymastaadmi/quiet-neurons**

THE DESIGN PRESSURE

A dense Transformer evaluates every feed-forward unit for every token; activation sparsity is the counter-proposal. But Transformer sparsity is already emergent [3], needs a hard rectifier for exact zeros [4], and is input-dependent in *which* units fire, not merely how many [5]. Sparsity alone distinguishes nothing. What distinguishes an architecture is what its sparsity depends on.

WHAT BDH CHANGES

Dragon Hatchling (BDH) [1] makes it structural. In the BDH-GPU formulation Pathway open-sources, every layer multiplies two rectified (ReLU) vectors elementwise, and a product of two sparse non-negative vectors is sparser than either. *Active* means a strictly non-zero entry in that product; a rectifier cannot emit a negative, so an active neuron is a positive amount of something rather than a signed coordinate, which lets it read as a concept. That product is the paper's $y_{t,l}$: Eq. (8) defines it as $(D_y LN(\rho x))_+ \odot x$, Appendix E's listing reads `relu(ln(a) @ decoder_y) * x`, and Fig. 14 counts its non-zeros. Mind the naming collision: the released code's `y_sparse` is the paper's rectified *intermediate*; the paper's y is the code's `xy_sparse`. Attention is the second change: query and key are the same tensor, so the score matrix is a causally-masked Gram matrix of activations with **no softmax**. Positions whose neurons fired alike bind.

THE CLAIM, AND THE CORRECTION IT NEEDED

Section 6.4 reports that "fewer neurons are active ... for more predictable input signals": at layer 2 of a 65,536-neuron model, 4.0–7.5% active while memorising a new word against about 2.5% while repeating it, a 1.6–3.0× ratio [1, Fig. 14]. We reproduced that protocol on three models trained here from the public code, all for 2,309 steps, and it survives a 32-fold shrink: **1.47×, 2.12× and 1.87×** at 2,048, 8,192 and 16,384 neurons, each 1,280 sequences over five independently seeded samples, spread ±0.005 half-width. **In aggregate our three models agree with the paper: predictable text is quieter.** What follows sharpens that rather than contradicting it.

The same measurement narrows what that result can mean. Two blocks of the same sequence are both predicted almost perfectly yet differ 2.65-fold in activity, over 1,280 sequences of the 8,192-neuron model: the 13-letter warm-up runs at **9.7%** of layer 2 active at 0.0004 nats of mean surprise (a nat is the natural-log unit of cross-entropy; zero is perfect), the 56 repeated letters at **3.6%** at 0.004 nats. The warm-up is identical in every training sequence, so it lives in the **weights**; the repeated word is new each time, so it can only come from **context**. What *covaries* with the difference is not predictability but **where the knowledge came from**. That is a correlation, not an established cause. A learner falsifies it in a minute: inject an unpredictable letter mid-repetition and activity rises there, while the letter before is unchanged to nine decimals, the model reading only leftwards.

System	Where its sparsity comes from	Carried per token	Evidence status
Dense Transformer feed-forward	Emergent in training [3]; needs a hard rectifier for exact zeros [4]	A key-value cache, growing with context	Extensive, independent
BDH-GPU, measured here	Structural: elementwise product of two rectified vectors, sparser than either	A fixed-size synaptic state	Reproduced here at toy scale; author-reported at 65,536 neurons
BDH-CQ	Not public	Not public	Developer-reported only

EVIDENCE, LABELLED

Reproduced here: the three ratios, the counterexample, the surprise-injection check, and parity of the in-browser model against PyTorch (logit max difference 6.1e-5, twelve of twelve sparsity series bit-identical) — an implementation check, not evidence for the architectural claim. **Developer-reported, not reproduced:** the Fig. 14 band; 97.4% on Sudoku Extreme, which Pathway's repository says comes from their internal implementation and does not reproduce from the open code; the 1B-to-600B scaling runs; BDH-CQ's ARC-AGI-1 result [2]. **Not modelled:** BDH-CQ's internals, "proprietary" by its authors' own statement [2]; **it has no role here**, and inventing one would be worse. Pathway's strongest external grounding is development on Amazon SageMaker HyperPod, a commercial partnership, not an independent evaluation. **Where it is likely worse:** sparse non-negative activations plausibly cost capacity against dense superposition, and no published comparison runs past GPT-2 scale.

LIMITATIONS, MATURITY, AND WHAT WOULD SETTLE THEM

Depth matters: layer 0 runs *backwards* at all three sizes, ratios below 1 (0.80, 0.80, 0.86) meaning *more* neurons fire while repeating; layer 3 is flat below 16,384. Scaling is **not monotonic** and we published the opposite before the third model finished. The wall-clock confound is gone, the short two retrained to 2,309 steps, dip surviving at **40**× the mean half-width; none finished the 4,000-step schedule, though retraining n=2,048 to completion moves layer 2 only 1.4705 to 1.4620. These are toys, 0.4M to 6.3M parameters on a synthetic task, as the paper's is. Surprise and sparsity do *not* track per letter (Pearson 0.30, Spearman −0.05), killing a stronger claim. Most importantly we establish the *signature*, not its cause. We intervened twice. Fixing the word collapses the gap 2.35× to 1.04×, but that control is degenerate and separates nothing. Drawing the word instead from a pool of 256 does separate them, letters in the weights and identity still from context: that model is **4.5× easier by loss (0.038 against 0.174) yet shows a larger effect, 2.54× against 2.35×**, which is the opposite of what a difficulty confound predicts. We registered in advance that the ratio would rise monotonically with pool size. **It did not**, and K=16 inverts to 0.92, unexplained. **Maturity:** credible and outsider-checkable for this narrow property; the wider programme is earlier, its headline results developer-reported.

[1] Kosowski, Uznański, Chorowski, Stamirowska, Bartoszkiewicz, *The Dragon Hatchling*, arXiv:2509.26507 (2025), §6.4 and Fig. 14. [2] Engdahl, Kosowski, Chorowski et al., *BDH-CQ*, arXiv:2608.09888 (2026). [3] Li, You, Bhojanapalli et al., *The Lazy Neuron Phenomenon*, arXiv:2210.06313 (2022), ICLR 2023. [4] Mirzadeh, Alizadeh, Mehta et al., *ReLU Strikes Back*, arXiv:2310.04564 (2023). [5] Liu, Wang, Dao et al., *Deja Vu*, arXiv:2310.17157 (2023), ICML 2023.

Where to continue, in order. [1] §6.4 and Fig. 14 for the result reproduced here, then [1] §3.2 and Eq. (8) for the equations defining what is counted, then [3] for why sparsity alone distinguishes no architecture, then the artifact for the counterexample, which no paper states. — Reproduce every number: **python measure.py --ckpt checkpoints/bdh_n2048.pt --embd 64 --mult 32 --repeats 5**. Code and data MIT; `bdh.py` is Pathway's, unmodified, MIT.